

EXERCISES

QUESTION 1:

Write a program that generates the following output by using setw, left, right and endl manipulators.

```
Programming
      Fundamentals
            C++
```

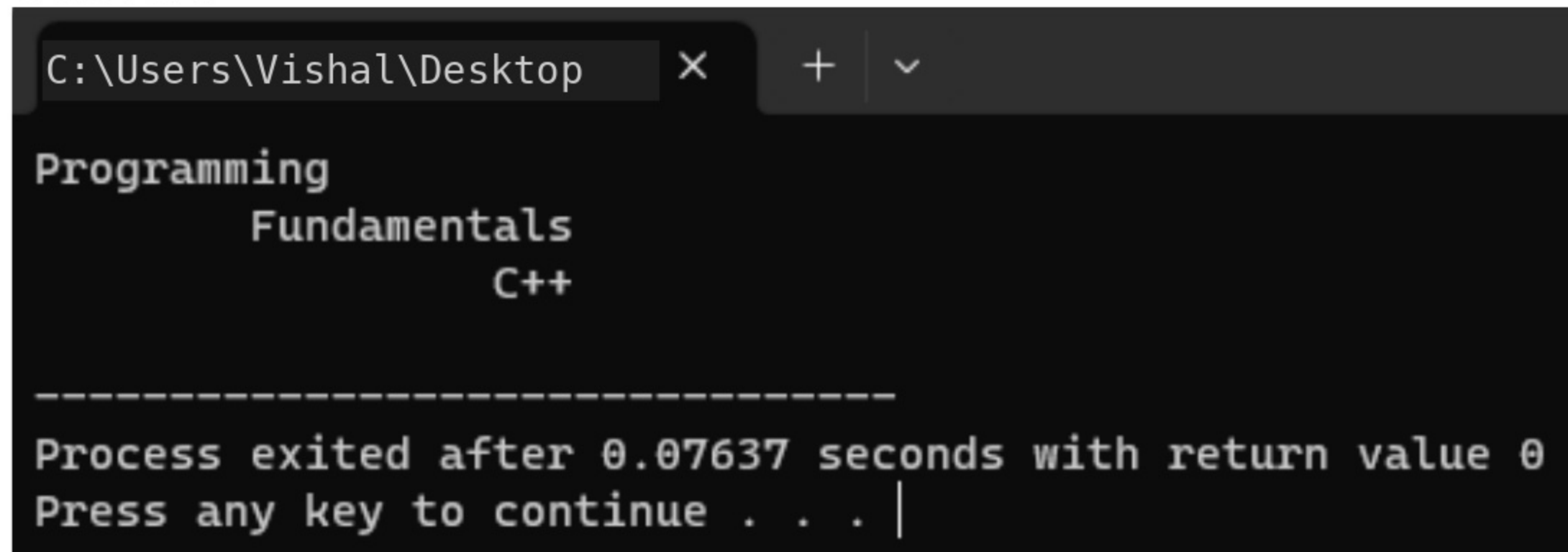
CODE:

```
#include <iostream>
#include <iomanip>
using namespace std;
int main(){

    cout<<"Programming" <<endl
        <<setw(20) <<"Fundamentals" <<endl
        <<setw(20) <<"C++" <<endl;

    return 0;
}
```

OUTPUT:



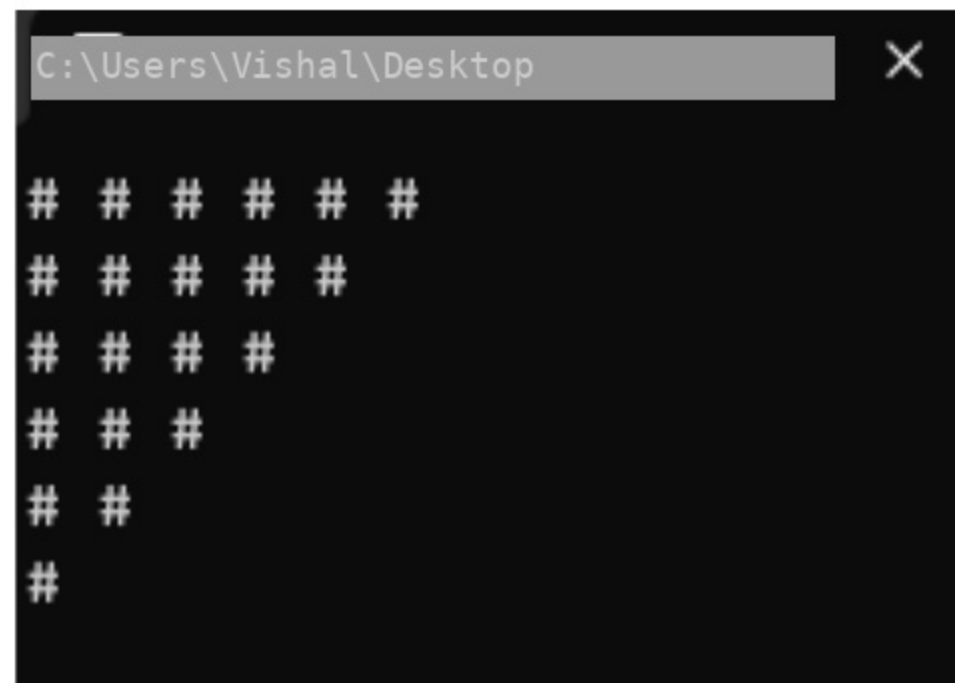
```
C:\Users\Vishal\Desktop  x  +  v
Programming
      Fundamentals
            C++

-----
Process exited after 0.07637 seconds with return value 0
Press any key to continue . . . |
```

QUESTION 2 (a):

Write a program to print the following shapes using the **setw** and **endl** manipulator only (Don't use and escape sequence or spaces).

Hint: setw only affects the immediately following output. It doesn't persist for subsequent outputs.



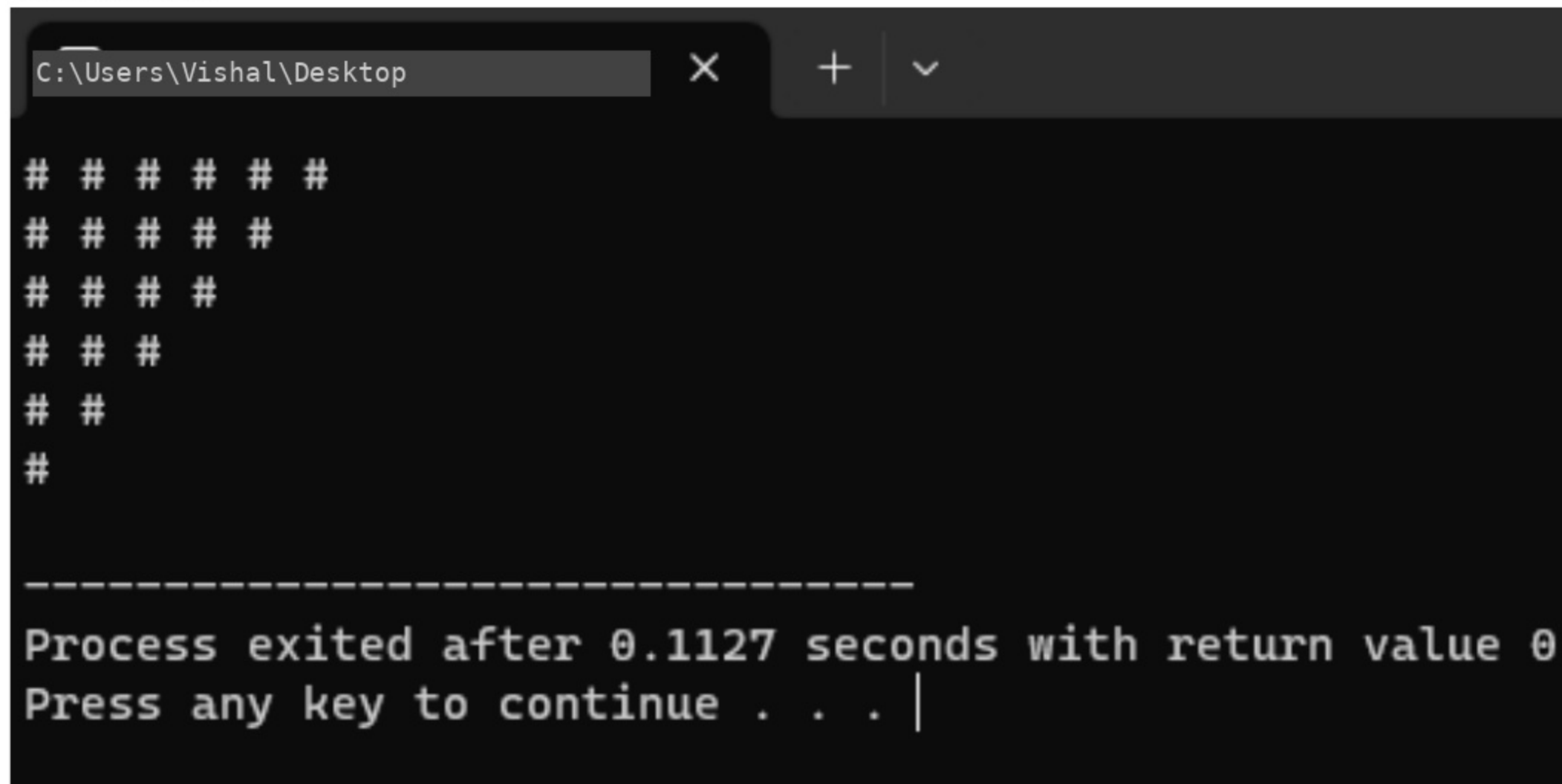
CODE:

```
#include <iostream>
#include <iomanip>
using namespace std;
int main(){

    cout << "#" << setw(2) << "#" << setw(2) << "#" << setw(2) << "#" << setw(2) << "#" <<
    setw(2) << "#" << endl;
    cout << "#" << setw(2) << "#" << setw(2) << "#" << setw(2) << "#" << setw(2) << "#" << endl;
    cout << "#" << setw(2) << "#" << setw(2) << "#" << setw(2) << "#" << endl;
    cout << "#" << setw(2) << "#" << setw(2) << "#" << endl;
    cout << "#" << setw(2) << "#" << endl;
    cout << "#" << endl;

    return 0;
}
```

OUTPUT:



QUESTION 2 (b):



CODE:

```
#include <iostream>
```

```
#include <iomanip>
```

```
using namespace std;
```

```
int main(){
```

```
    cout << "#" << endl;
```

```
    cout << "#" << setw(2) << "#" << endl;
```

```
    cout << "#" << setw(2) << "#" << setw(2) << "#" << endl;
```

```
    cout << "#" << setw(2) << "#" << setw(2) << "#" << setw(2) << "#" << endl;
```

```
    cout << "#" << setw(2) << "#" << setw(2) << "#" << setw(2) << "#" << setw(2) << "#" << endl;
```

```
    cout << "#" << setw(2) << "#" << setw(2) << "#" << setw(2) << "#" << endl;
```

```
    cout << "#" << setw(2) << "#" << setw(2) << "#" << endl;
```

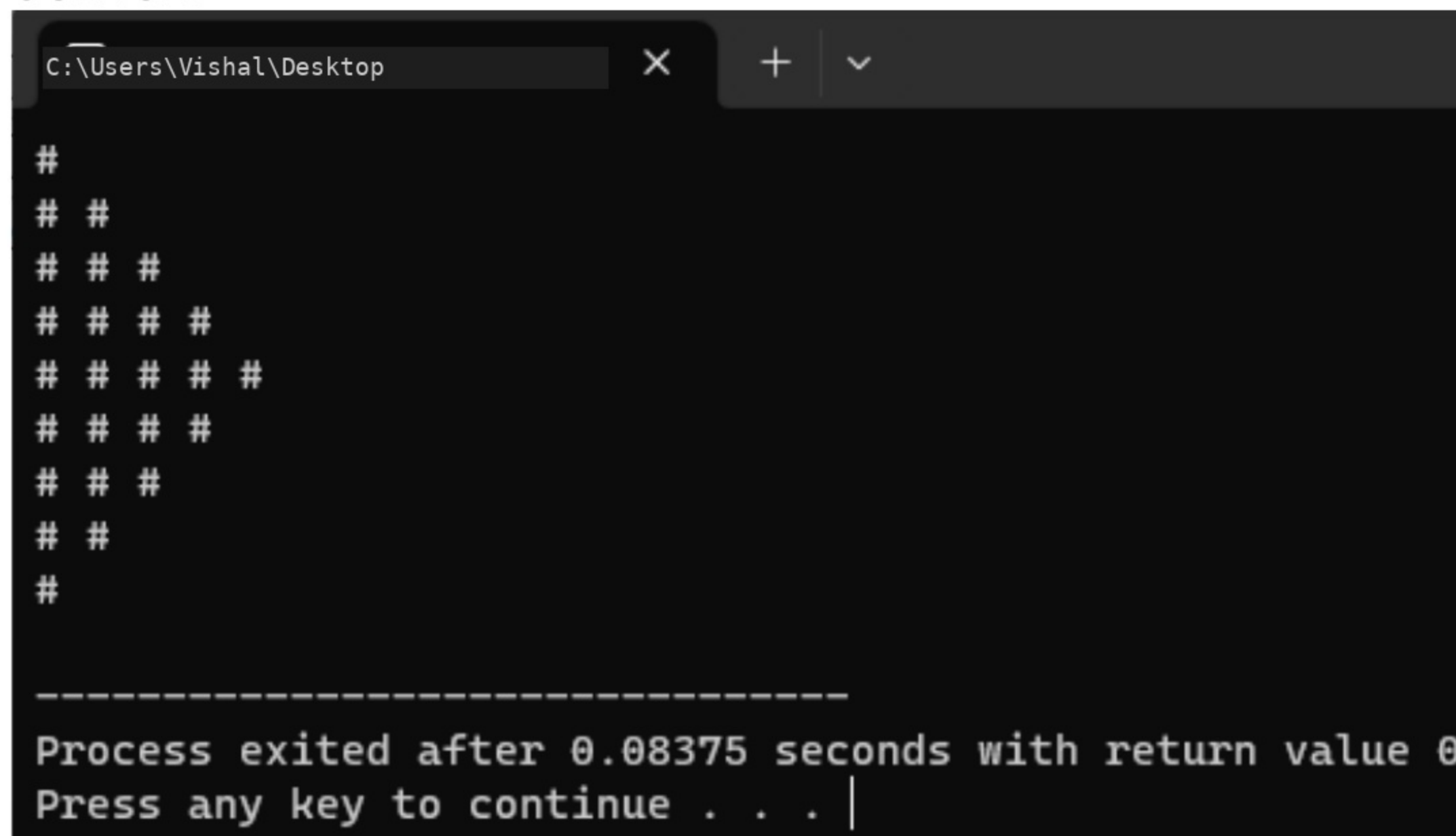
```
    cout << "#" << setw(2) << "#" << endl;
```

```
    cout << "#" << endl;
```

```
return 0;
```

```
}
```

OUTPUT:



QUESTION 2 (c):

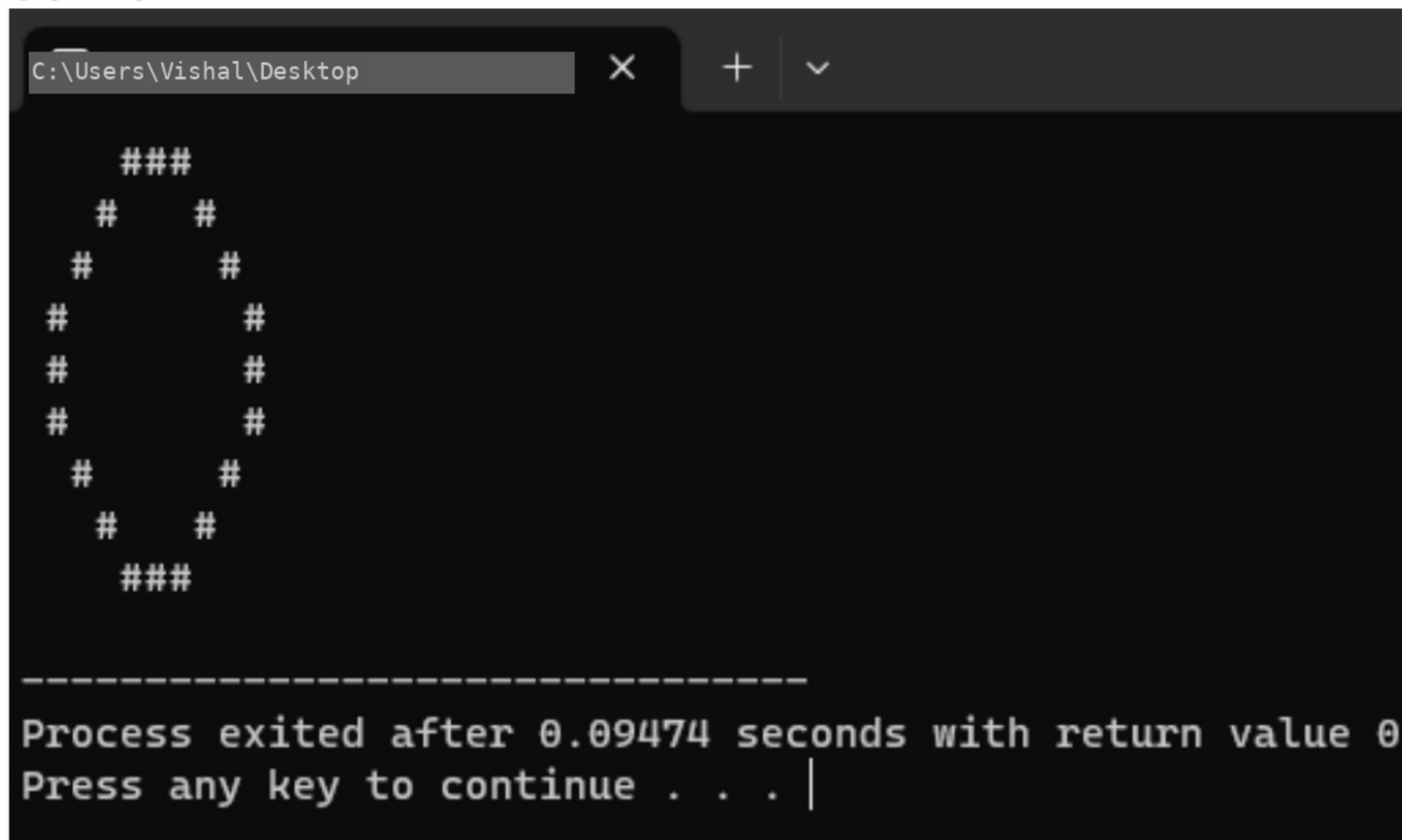


CODE:

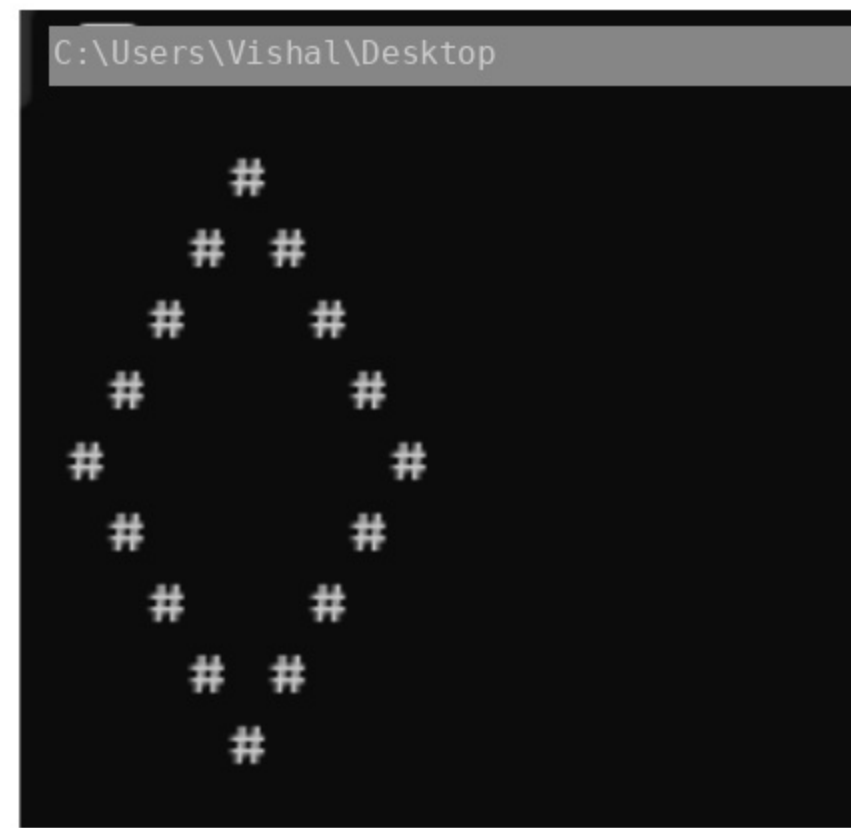
```
#include<iostream>
#include<iomanip>
using namespace std;
int main()
{
    cout<<setw(7)<<"###"<<endl;
    cout<<setw(4)<<"#"<<setw(4)<<"#"<<endl;
    cout<<setw(3)<<"#"<<setw(6)<<"#"<<endl;
    cout<<setw(2)<<"#"<<setw(8)<<"#"<<endl;
    cout<<setw(2)<<"#"<<setw(8)<<"#"<<endl;
    cout<<setw(2)<<"#"<<setw(8)<<"#"<<endl;
    cout<<setw(3)<<"#"<<setw(6)<<"#"<<endl;
    cout<<setw(4)<<"#"<<setw(4)<<"#"<<endl;
    cout<<setw(7)<<"###"<<endl;

    return 0;
}
```

OUTPUT:



QUESTION 2 (d):



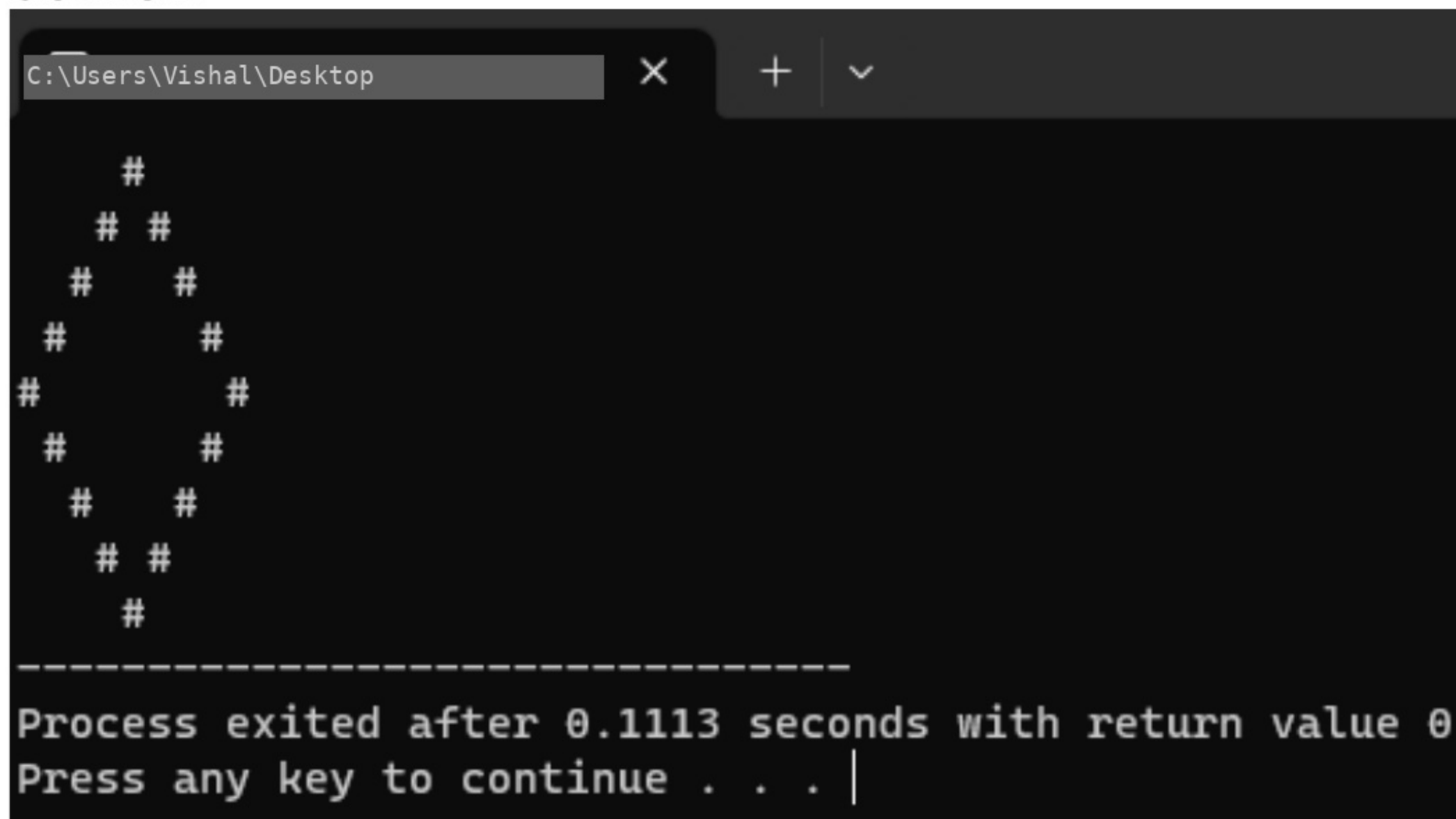
CODE:

```
#include <iostream>
#include <iomanip>
using namespace std;
int main(){

    cout << setw(5) << "#" << endl;
    cout << setw(4) << "#" << setw(2) << "#" << endl;
    cout << setw(3) << "#" << setw(4) << "#" << endl;
    cout << setw(2) << "#" << setw(6) << "#" << endl;
    cout << setw(1) << "#" << setw(8) << "#" << endl;
    cout << setw(2) << "#" << setw(6) << "#" << endl;
    cout << setw(3) << "#" << setw(4) << "#" << endl;
    cout << setw(4) << "#" << setw(2) << "#" << endl;
    cout << setw(5) << "#";

    return 0;
}
```

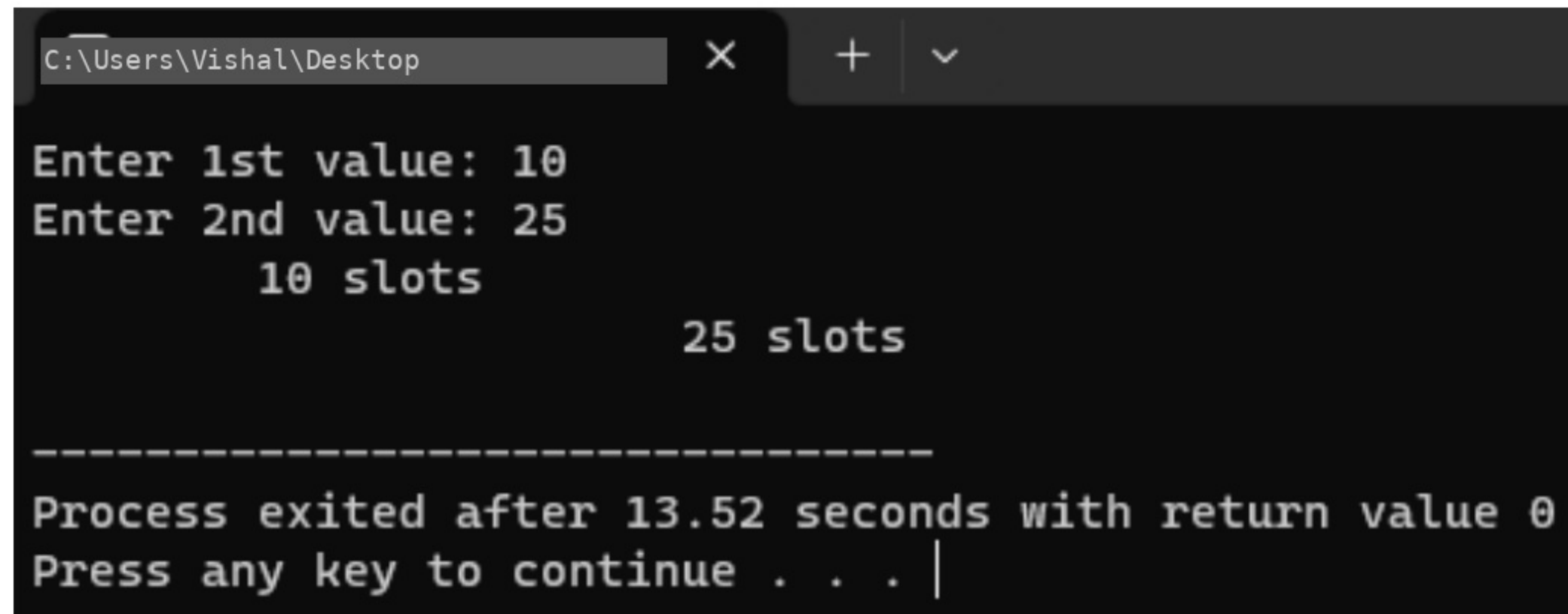
OUTPUT:



QUESTION 3:

Write a program that asks the user to input two numbers i.e., num1 and num2;

And the output should be formatted according to these numbers. The larger number will print the larger gap on the screen (use setw and right align).



```
C:\Users\Vishal\Desktop X + v
Enter 1st value: 10
Enter 2nd value: 25
    10 slots
                25 slots

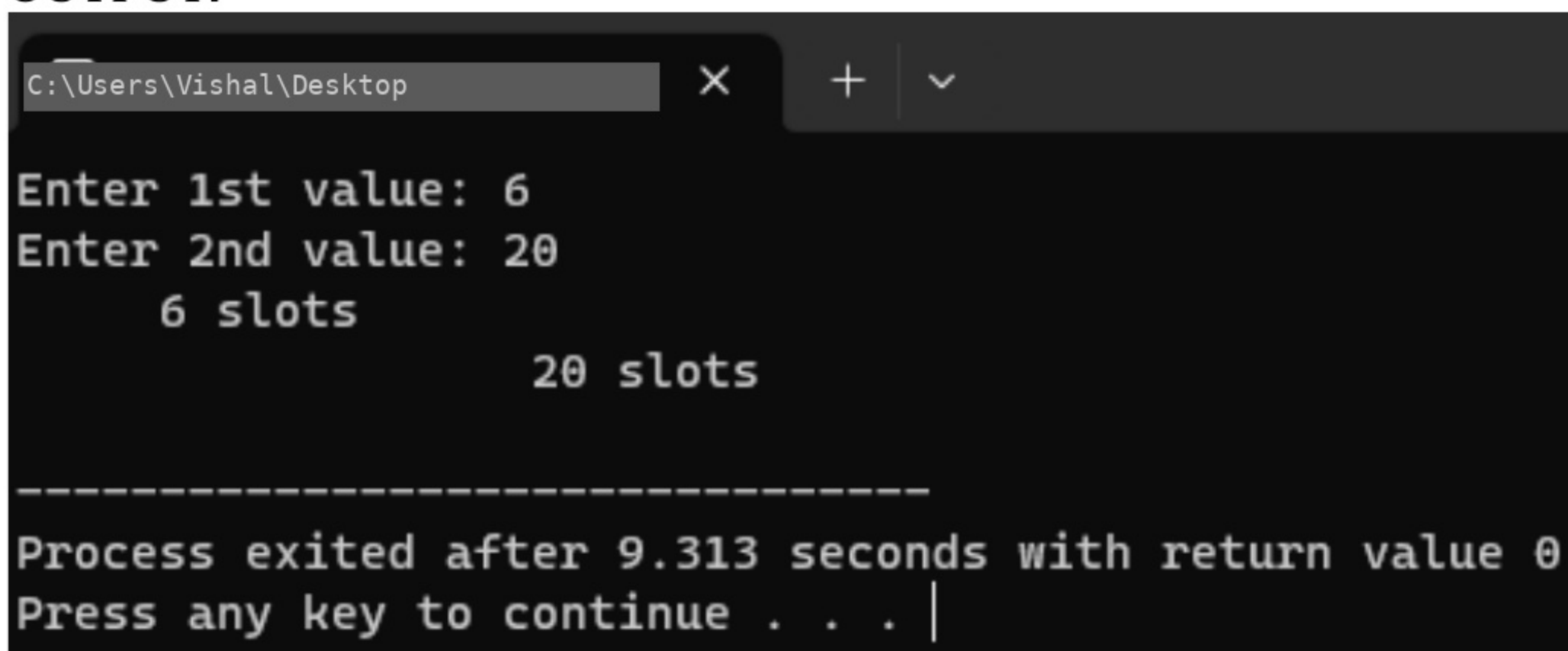
-----
Process exited after 13.52 seconds with return value 0
Press any key to continue . . . |
```

CODE:

```
#include <iostream>
#include <iomanip>
using namespace std;

int main(){
    int num1,num2;
    cout<<"Enter 1st value: ";
    cin>>num1;
    cout<<"Enter 2nd value: ";
    cin>>num2;
    cout<<setw(num1)<<num1<<" slots"<<endl
         <<setw(num2)<<num2<<" slots"<<endl;
    return 0;
}
```

OUTPUT:



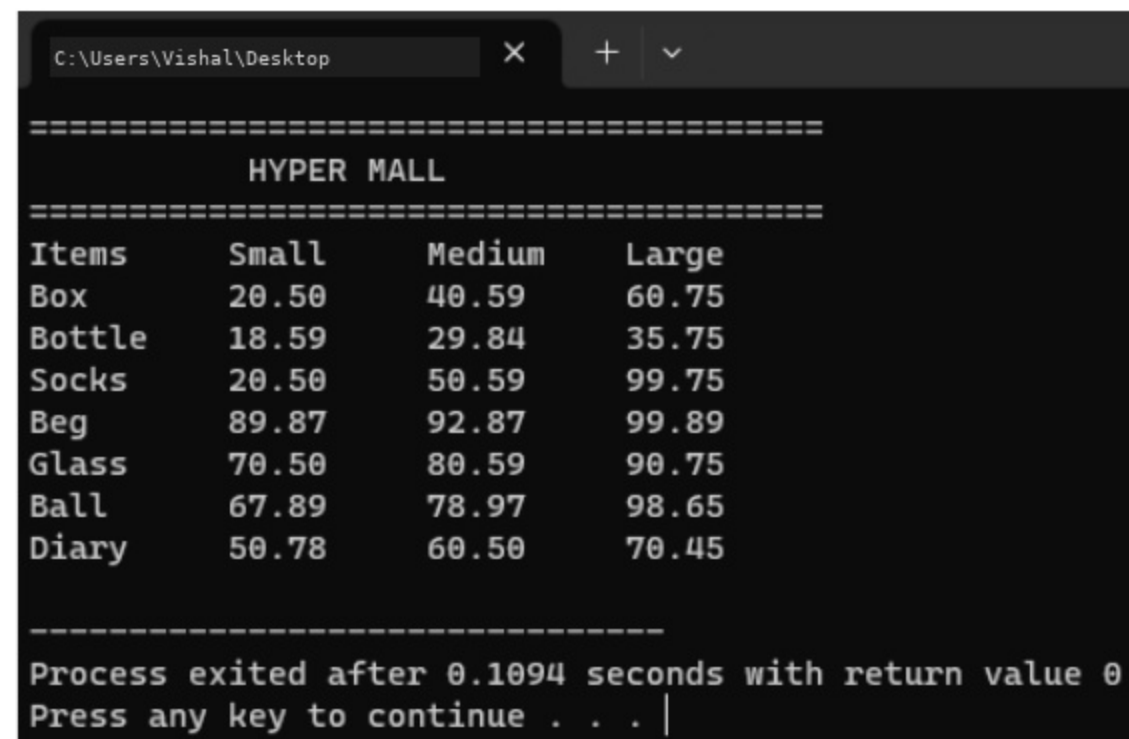
```
C:\Users\Vishal\Desktop X + v
Enter 1st value: 6
Enter 2nd value: 20
    6 slots
                20 slots

-----
Process exited after 9.313 seconds with return value 0
Press any key to continue . . . |
```


QUESTION 4:

Generate the below-mentioned output using the Manipulators only (setw(n), endl, fixed, left, etc.)

Hint: You may use setprecision(n), left/right, and fixed only once in a program. No need to use them again and again.

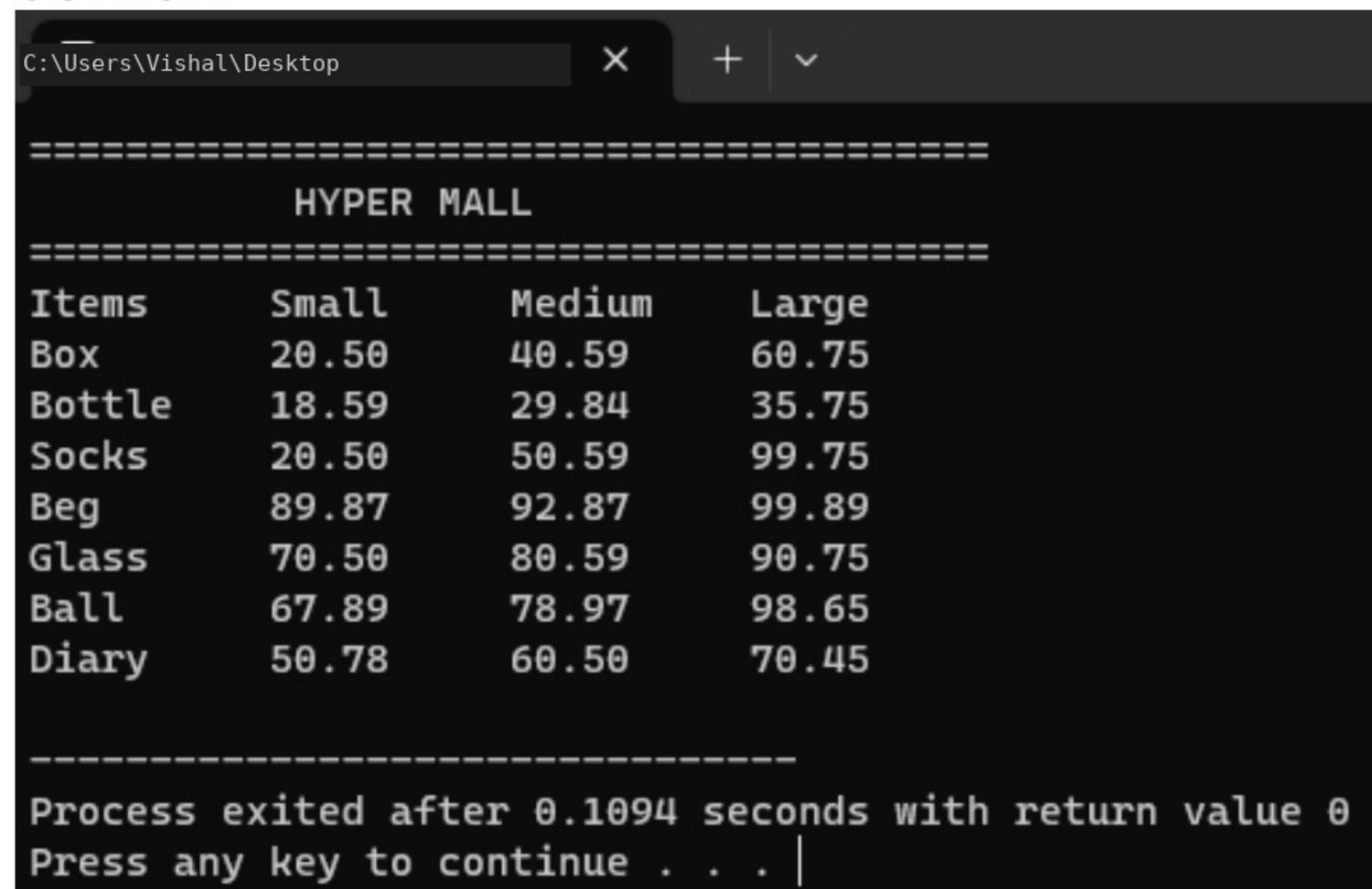


```
=====
                        HYPER MALL
=====
Items      Small      Medium      Large
Box         20.50       40.59       60.75
Bottle      18.59       29.84       35.75
Socks       20.50       50.59       99.75
Beg         89.87       92.87       99.89
Glass       70.50       80.59       90.75
Ball        67.89       78.97       98.65
Diary       50.78       60.50       70.45
=====
Process exited after 0.1094 seconds with return value 0
Press any key to continue . . . |
```

CODE:

```
#include <iostream>
#include <iomanip>
using namespace std;
int main(){
cout<<"=====\n";
cout<<"\t HYPER MALL\n";
cout<<"=====\n";
cout<<left<<setw(10)<<"Items"<<setw(10)<<"Small"<<setw(10)<<"Medium"<<setw(10)<<"Large"<<
endl;
cout<<setprecision(2)<<fixed
    <<setw(10)<<"Box"<<setw(10)<<20.50<<setw(10)<<40.59<<setw(10)<<60.75<<endl;
cout<<setw(10)<<"Bottle"<<setw(10)<<18.59<<setw(10)<<29.84<<setw(10)<<35.75<<endl;
cout<<setw(10)<<"Socks"<<setw(10)<<20.50<<setw(10)<<50.59<<setw(10)<<99.75<<endl;
cout<<setw(10)<<"Beg"<<setw(10)<<89.87<<setw(10)<<92.87<<setw(10)<<99.89<<endl;
cout<<setw(10)<<"Glass"<<setw(10)<<70.50<<setw(10)<<80.59<<setw(10)<<90.75<<endl;
cout<<setw(10)<<"Ball"<<setw(10)<<67.89<<setw(10)<<78.97<<setw(10)<<98.65<<endl;
cout<<setw(10)<<"Diary"<<setw(10)<<50.78<<setw(10)<<60.50<<setw(10)<<70.45<<endl;
return 0;
}
```

OUTPUT:

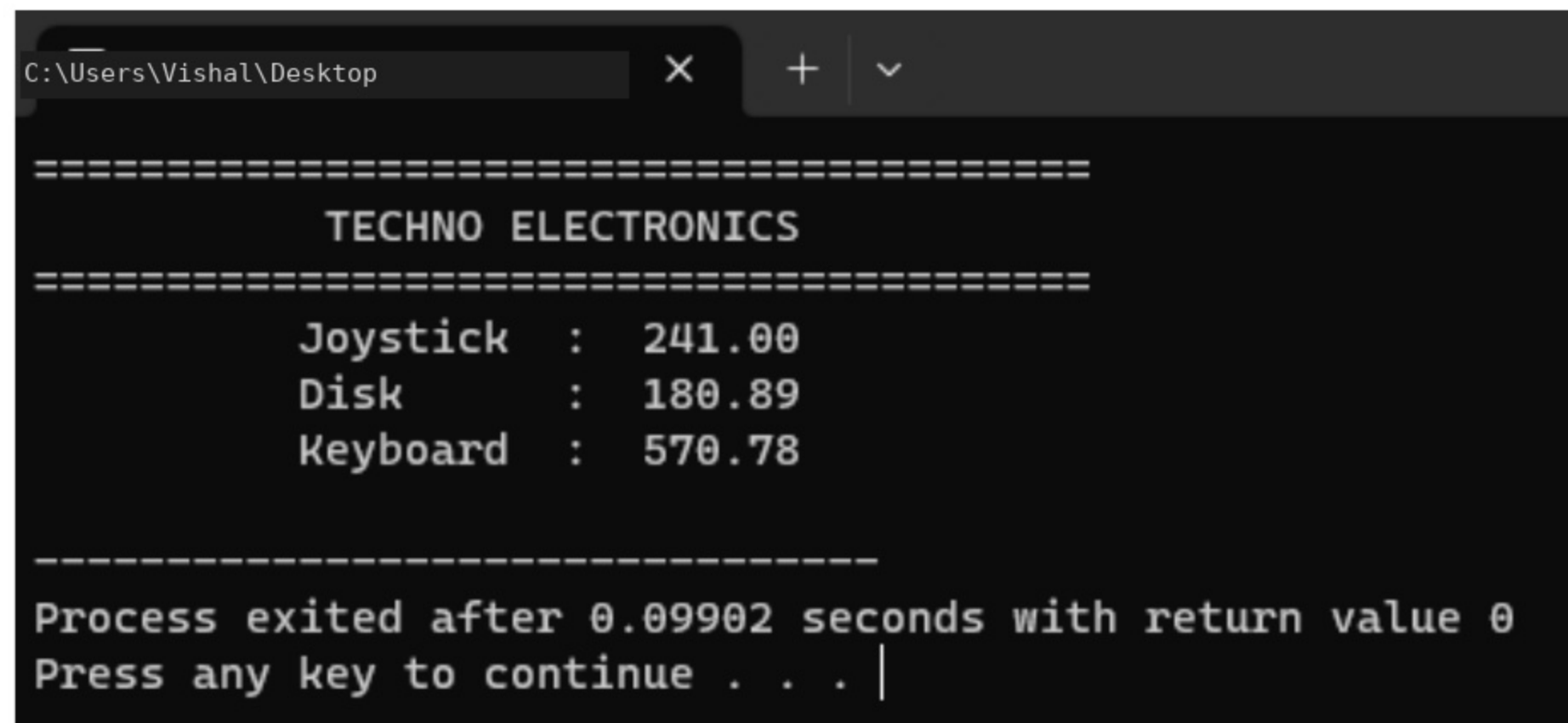


```
=====
                        HYPER MALL
=====
Items      Small      Medium      Large
Box         20.50       40.59       60.75
Bottle      18.59       29.84       35.75
Socks       20.50       50.59       99.75
Beg         89.87       92.87       99.89
Glass       70.50       80.59       90.75
Ball        67.89       78.97       98.65
Diary       50.78       60.50       70.45
=====
Process exited after 0.1094 seconds with return value 0
Press any key to continue . . . |
```

QUESTION 5:

Write a program in which you create three float variables to set the price of 240.99999 for a joystick, 180.88888 for disk, and 570.77777 for keyboard. Your task is to print the prices of the above-mentioned items in the following format.

NOTE: After decimal points, your program must print two digits.



```
=====
                TECHNO ELECTRONICS
=====
Joystick   :   241.00
Disk       :   180.89
Keyboard   :   570.78

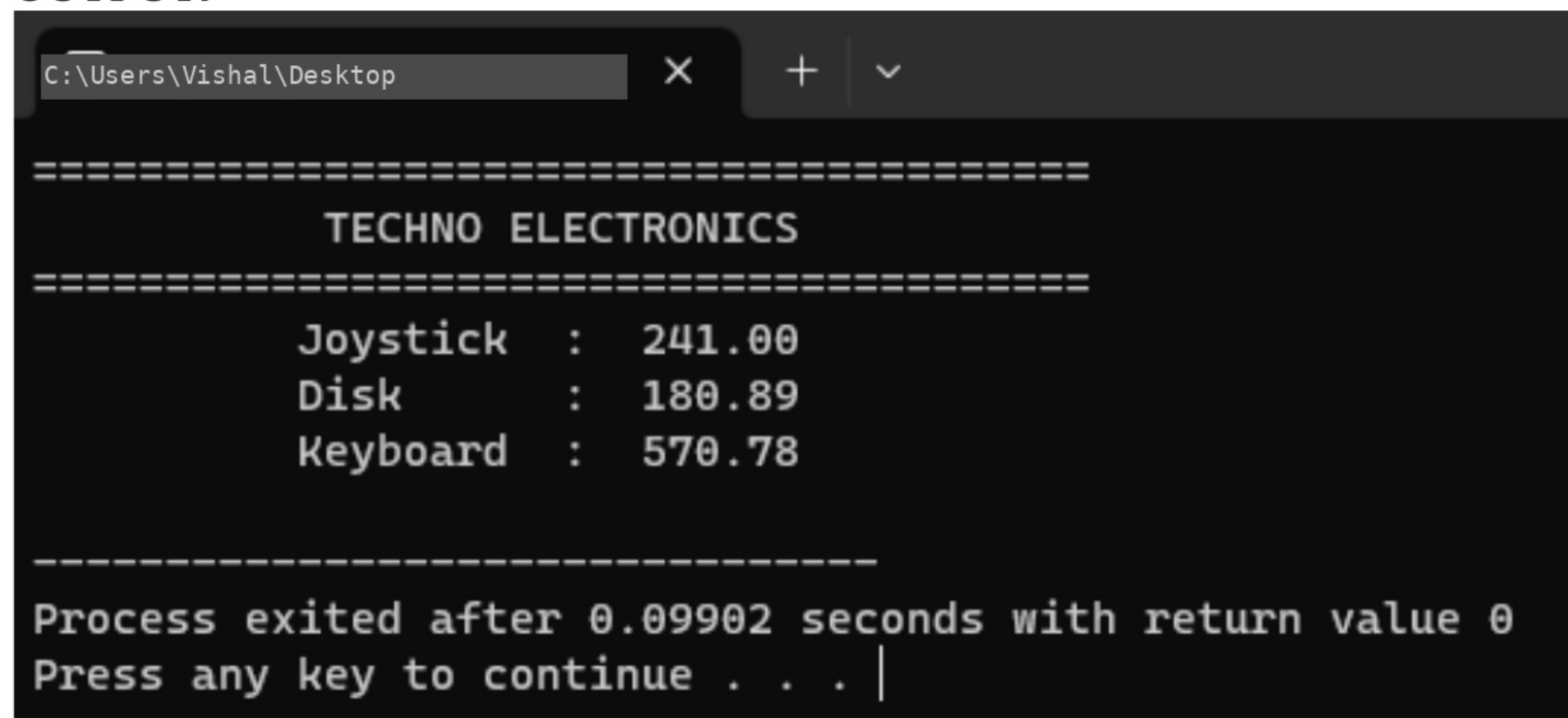
-----
Process exited after 0.09902 seconds with return value 0
Press any key to continue . . . |
```

CODE:

```
#include <iostream>
#include <iomanip>
using namespace std;

int main(){
    float num1=240.99999,num2=180.88888,num3=570.77777;
    cout<<"=====\\n";
    cout<<"\\t TECHNO ELECTRONICS\\n";
    cout<<"=====\\n";
    cout<<setw(10)<<"
    "<<left<<setw(10)<<"Joystick"<<setw(3)<<":"<<setw(8)<<setprecision(2)<<fixed<<num1<<endl;
    cout<<setw(10)<<"
    "<<left<<setw(10)<<"Disk"<<setw(3)<<":"<<setw(8)<<setprecision(2)<<fixed<<num2<<endl;
    cout<<setw(10)<<"
    "<<left<<setw(10)<<"Keyboard"<<setw(3)<<":"<<setw(8)<<setprecision(2)<<fixed<<num3<<endl;
    return 0;
}
```

OUTPUT:



```
=====
                TECHNO ELECTRONICS
=====
Joystick   :   241.00
Disk       :   180.89
Keyboard   :   570.78

-----
Process exited after 0.09902 seconds with return value 0
Press any key to continue . . . |
```


QUESTION 6:

Write a program for the marksheet, where user should be able to enter the marks for five subjects. Then, your program should tell him/her his/her obtained marks, average marks and percentage.

Note: You can assume each subject to be of 100 marks. So, the total marks would become 500 for five subjects. While printing the final Marksheet it must contain your name and CMS-ID as student name and CMS-ID.

The sample output is shown below (This is just a sample; you may even come up with new ideas to make marksheet look more eye-catching)

```
C:\Users\Vishal\Desktop X + v

Enter marks of FP: 89.7
Enter marks of ICT: 76.97
Enter marks of OOP: 92.256
Enter marks of Maths: 78.65
Enter marks of English: 87.2
*****
                MARKSHEET
*****
Student Name:      Ayaz Ahmed
CMS ID:            222-22-2222
*****
SUBJECT            MARKS
*****
FP                  89.70
ICT                  76.97
OOP                  92.26
Maths                78.65
English              87.20
*****
Obt. Marks          424.78
Average              84.96
Percentage           84.96 %
*****

-----
Process exited after 51.85 seconds with return value 0
Press any key to continue . . . |
```

CODE:

```
#include <iostream>
#include <iomanip>
using namespace std;

int main(){
float a,b,c,d,e,obt;
float avg,per;
    cout<<"Enter marks of FP: ";
    cin>>a;
    cout<<"Enter marks of ICT: ";
    cin>>b;
    cout<<"Enter marks of OOP: ";
    cin>>c;
    cout<<"Enter marks of Maths: ";
    cin>>d;
    cout<<"Enter marks of English: ";
    cin>>e;

    obt=a+b+c+d+e;
    avg = (a+b+c+d+e)/5;
    per=obt/500*100;

    cout<<"*****\n";
    cout<<"\t MARKSHEET\n";
    cout<<"*****\n";

    cout<<left<<setw(20)<<"Student Name:"<<setw(10)<<"Ayaz Ahmed"<<endl;
    cout<<left<<setw(20)<<"CMS ID:"<<setw(10)<<"222-22-2222"<<endl;

    cout<<"*****\n";
    cout<<left<<setw(15)<<"SUBJECT"<<setw(7)<<"MARKS"<<endl;
    cout<<"*****\n";

    cout<<setprecision(2)<<fixed<<showpoint
        <<setw(15)<<"FP"<<setw(7)<<a<<endl;
    cout<<setw(15)<<"ICT"<<setw(7)<<b<<endl;
    cout<<setw(15)<<"OOP"<<setw(7)<<c<<endl;
    cout<<setw(15)<<"Maths"<<setw(7)<<d<<endl;
    cout<<setw(15)<<"English"<<setw(7)<<e<<endl;

    cout<<"*****\n";
    cout<<setw(15)<<"Obt. Marks"<<setw(7)<<obt<<endl;
    cout<<setw(15)<<"Average"<<setw(7)<<avg<<endl;
    cout<<setw(15)<<"Percentage"<<setw(7)<<per<<"%"<<endl;
    cout<<"*****\n";

    return 0;
}
```


OUTPUT:

```
C:\Users\Vishal\Desktop X + v

Enter marks of FP: 89.5
Enter marks of ICT: 78.8
Enter marks of OOP: 90.75
Enter marks of Maths: 92.5
Enter marks of English: 78.25
*****
                MARKSHEET
*****
Student Name:      Ayaz Ahmed
CMS ID:            222-22-2222
*****
SUBJECT            MARKS
*****
FP                  89.50
ICT                  78.80
OOP                  90.75
Maths                92.50
English              78.25
*****
Obt. Marks          429.80
Average              85.96
Percentage           85.96 %
*****

-----
Process exited after 50.84 seconds with return value 0
Press any key to continue . . . |
```